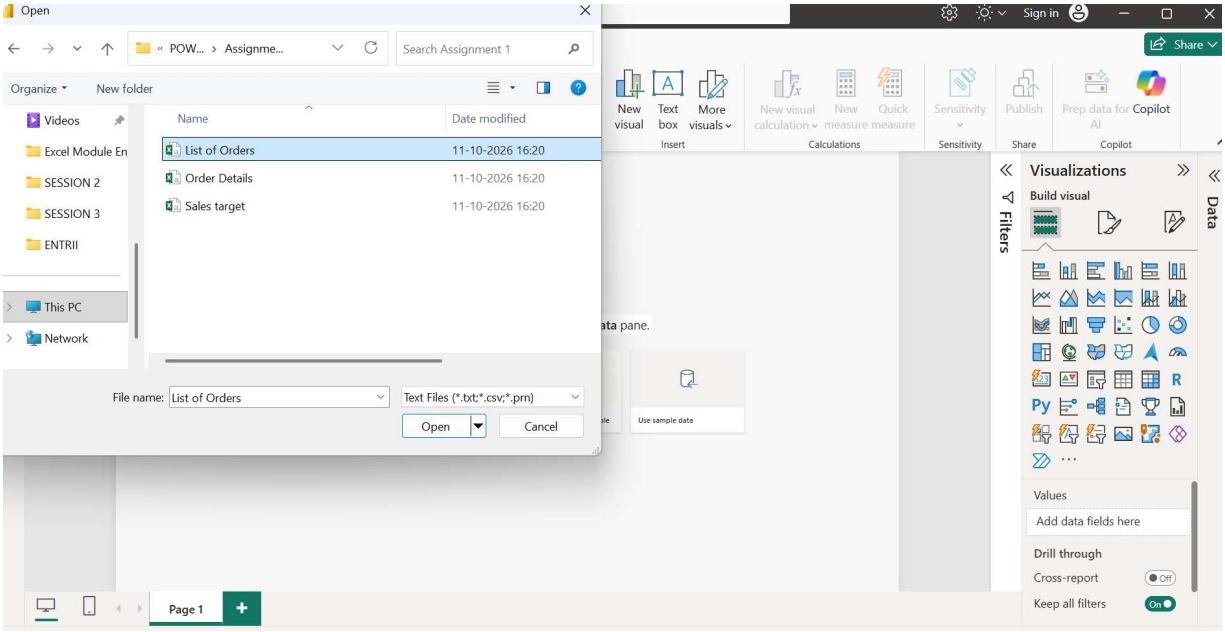
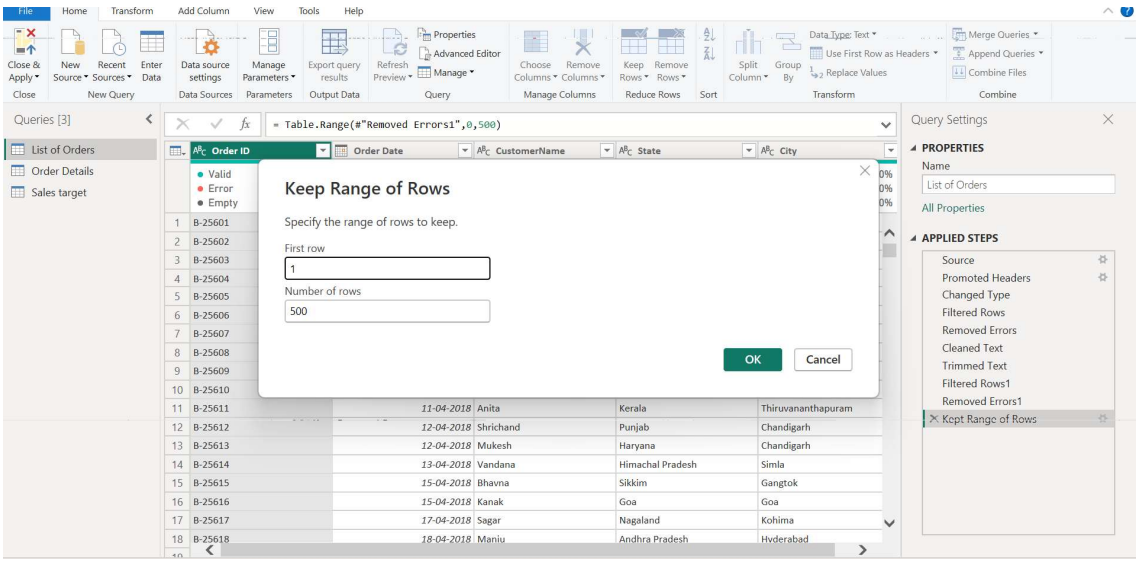
Data Import & Transformation

1.Data Import:

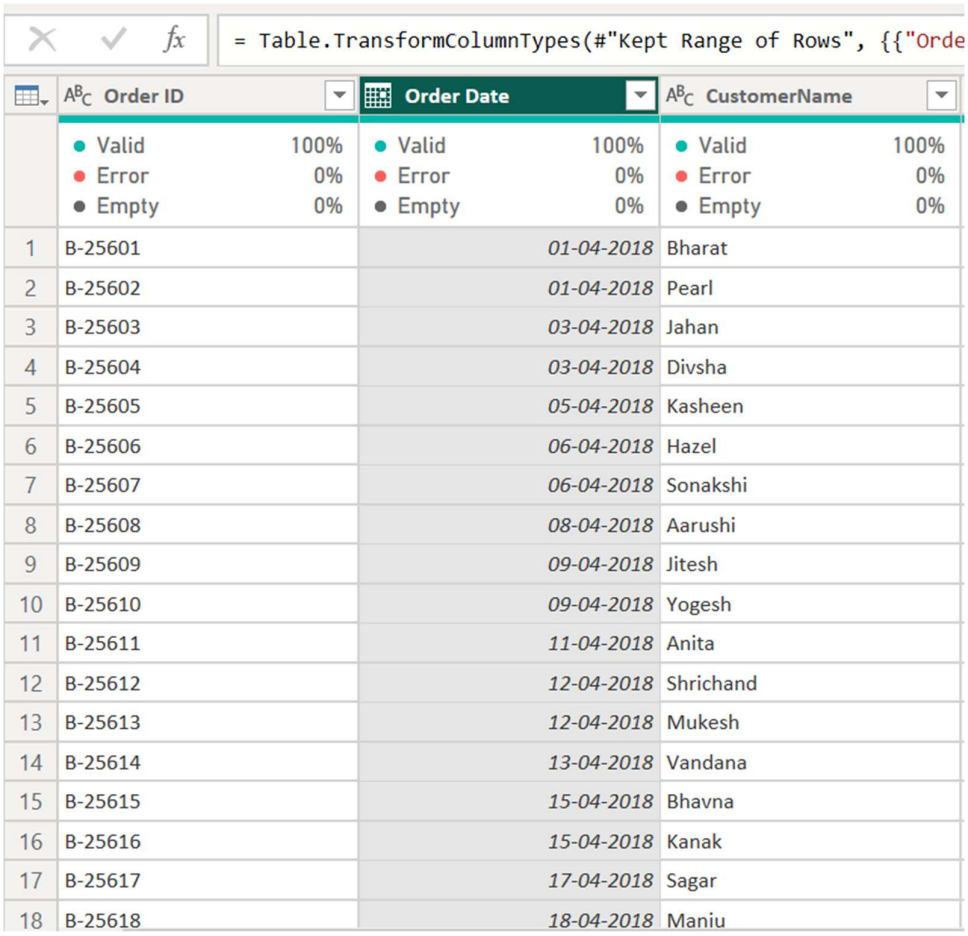


2.Row Limitting & Data Types:

- Restrict the List of Orders table to the first 500 rows.



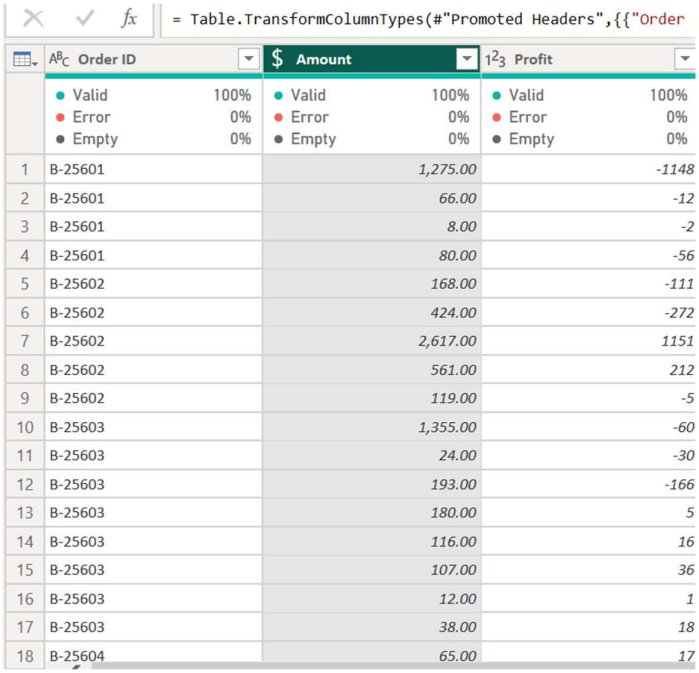
- Set the Order Date column to the Date data type.



= Table.TransformColumnTypes(#"Kept Range of Rows", {{"Orde

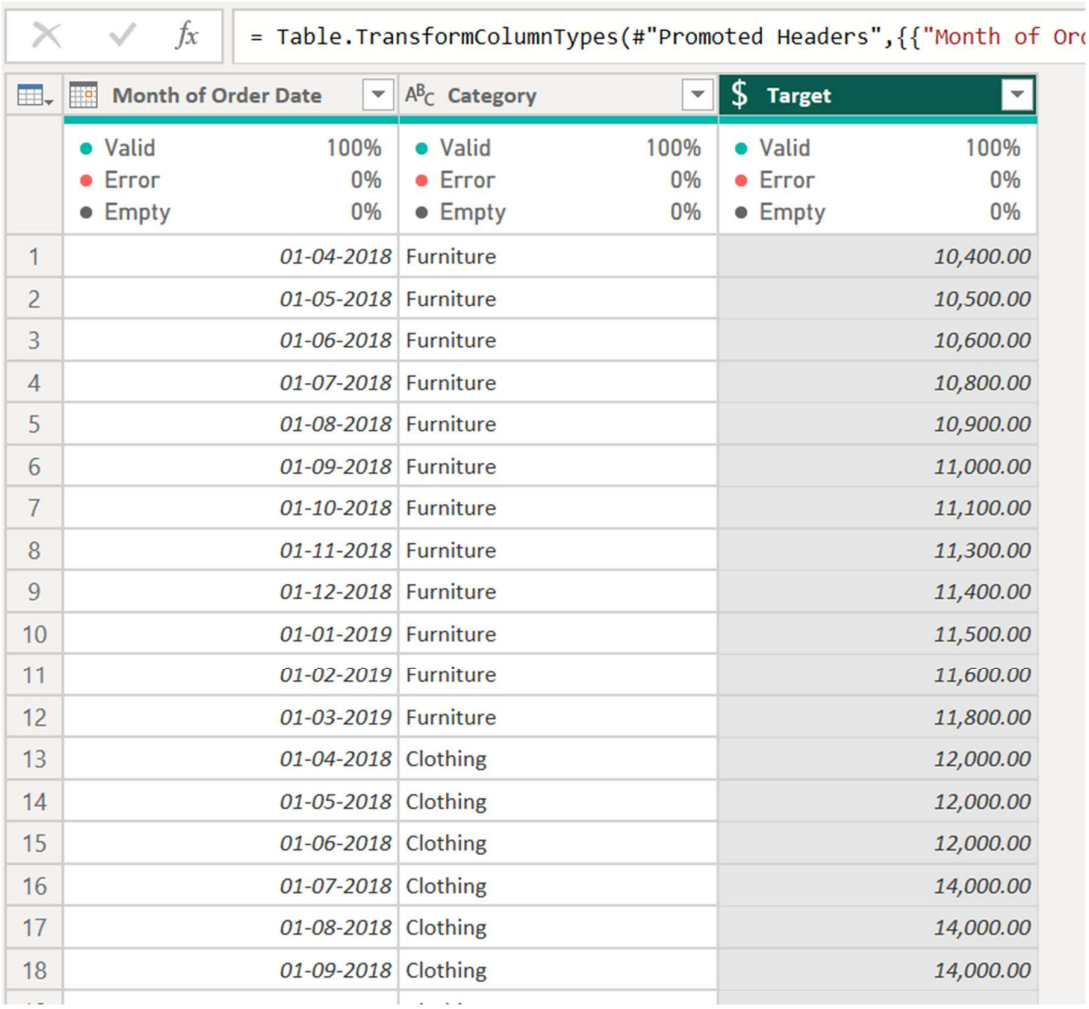
	Order ID	Order Date	CustomerName
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	B-25601	01-04-2018	Bharat
2	B-25602	01-04-2018	Pearl
3	B-25603	03-04-2018	Jahan
4	B-25604	03-04-2018	Divsha
5	B-25605	05-04-2018	Kasheen
6	B-25606	06-04-2018	Hazel
7	B-25607	06-04-2018	Sonakshi
8	B-25608	08-04-2018	Aarushi
9	B-25609	09-04-2018	Jitesh
10	B-25610	09-04-2018	Yogesh
11	B-25611	11-04-2018	Anita
12	B-25612	12-04-2018	Shrichand
13	B-25613	12-04-2018	Mukesh
14	B-25614	13-04-2018	Vandana
15	B-25615	15-04-2018	Bhavna
16	B-25616	15-04-2018	Kanak
17	B-25617	17-04-2018	Sagar
18	B-25618	18-04-2018	Maniu

- Change the Amount and Target columns to fixed decimal numbers.



= Table.TransformColumnTypes(#"Promoted Headers",{{"Order

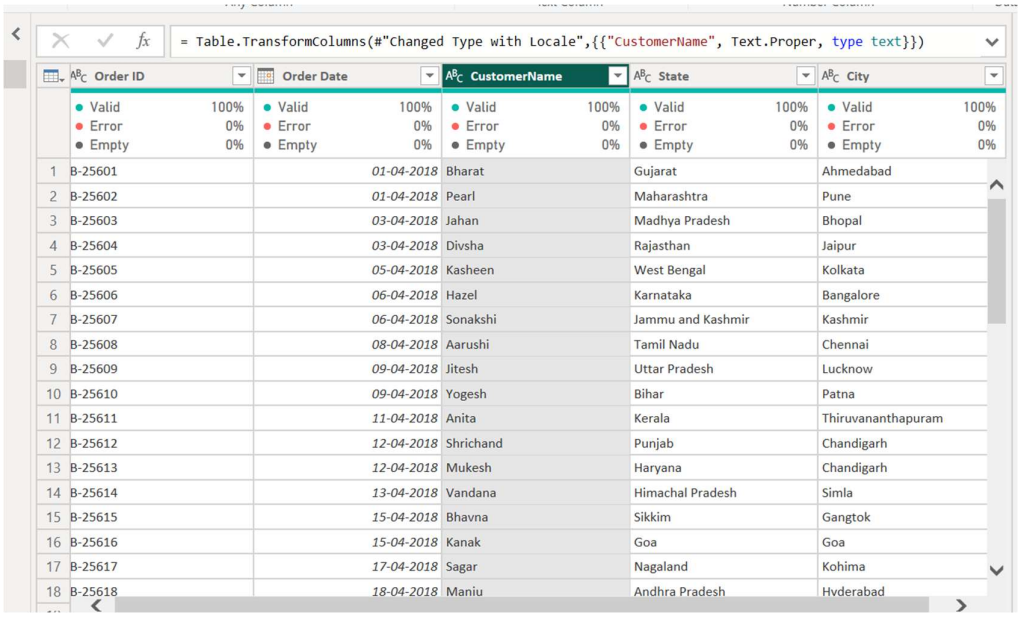
	Order ID	Amount	Profit
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	B-25601	1,275.00	-1148
2	B-25601	66.00	-12
3	B-25601	8.00	-2
4	B-25601	80.00	-56
5	B-25602	168.00	-111
6	B-25602	424.00	-272
7	B-25602	2,617.00	1151
8	B-25602	561.00	212
9	B-25602	119.00	-5
10	B-25603	1,355.00	-60
11	B-25603	24.00	-30
12	B-25603	193.00	-166
13	B-25603	180.00	5
14	B-25603	116.00	16
15	B-25603	107.00	36
16	B-25603	12.00	1
17	B-25603	38.00	18
18	B-25604	65.00	17



	Month of Order Date	Category	Target
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	01-04-2018	Furniture	10,400.00
2	01-05-2018	Furniture	10,500.00
3	01-06-2018	Furniture	10,600.00
4	01-07-2018	Furniture	10,800.00
5	01-08-2018	Furniture	10,900.00
6	01-09-2018	Furniture	11,000.00
7	01-10-2018	Furniture	11,100.00
8	01-11-2018	Furniture	11,300.00
9	01-12-2018	Furniture	11,400.00
10	01-01-2019	Furniture	11,500.00
11	01-02-2019	Furniture	11,600.00
12	01-03-2019	Furniture	11,800.00
13	01-04-2018	Clothing	12,000.00
14	01-05-2018	Clothing	12,000.00
15	01-06-2018	Clothing	12,000.00
16	01-07-2018	Clothing	14,000.00
17	01-08-2018	Clothing	14,000.00
18	01-09-2018	Clothing	14,000.00

3.Text Formatting:

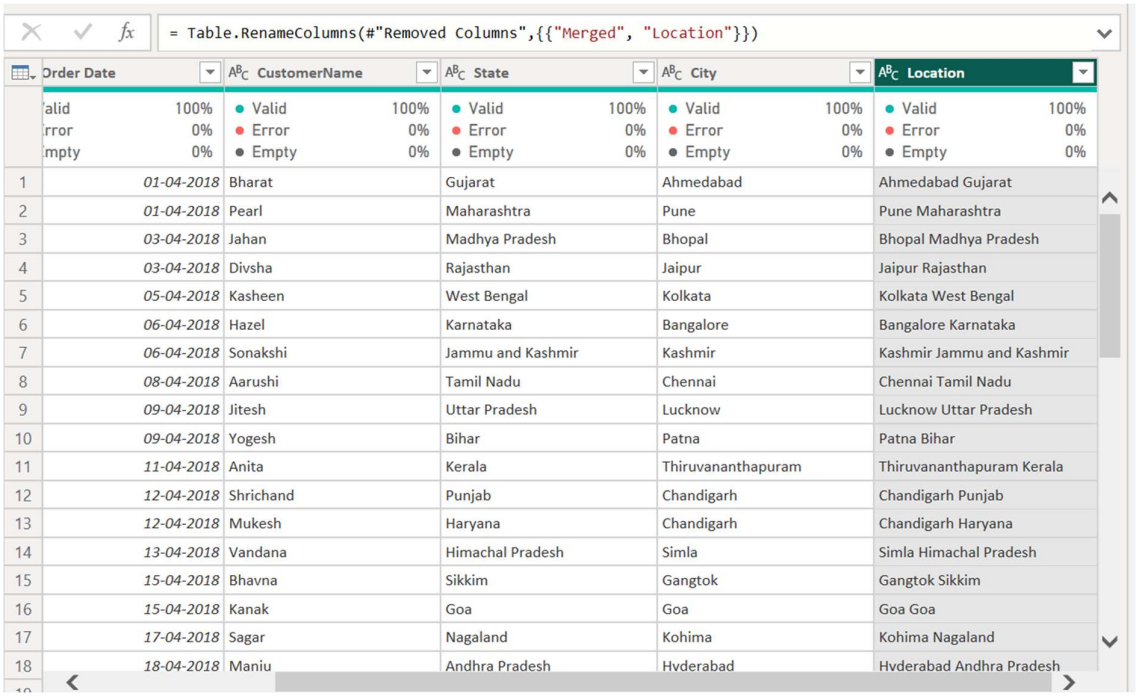
- Format the CustomerName column to proper case to ensure consistent capitalization.



	Order ID	Order Date	CustomerName	State	City
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
2	B-25602	01-04-2018	Pearl	Maharashtra	Pune
3	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
4	B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
5	B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
6	B-25606	06-04-2018	Hazel	Karnataka	Bangalore
7	B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
8	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
9	B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
10	B-25610	09-04-2018	Yogesh	Bihar	Patna
11	B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
12	B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
13	B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
14	B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
15	B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
16	B-25616	15-04-2018	Kanak	Goa	Goa
17	B-25617	17-04-2018	Sagar	Nagaland	Kohima
18	B-25618	18-04-2018	Maniu	Andhra Pradesh	Hyderabad

4.Column Creation:

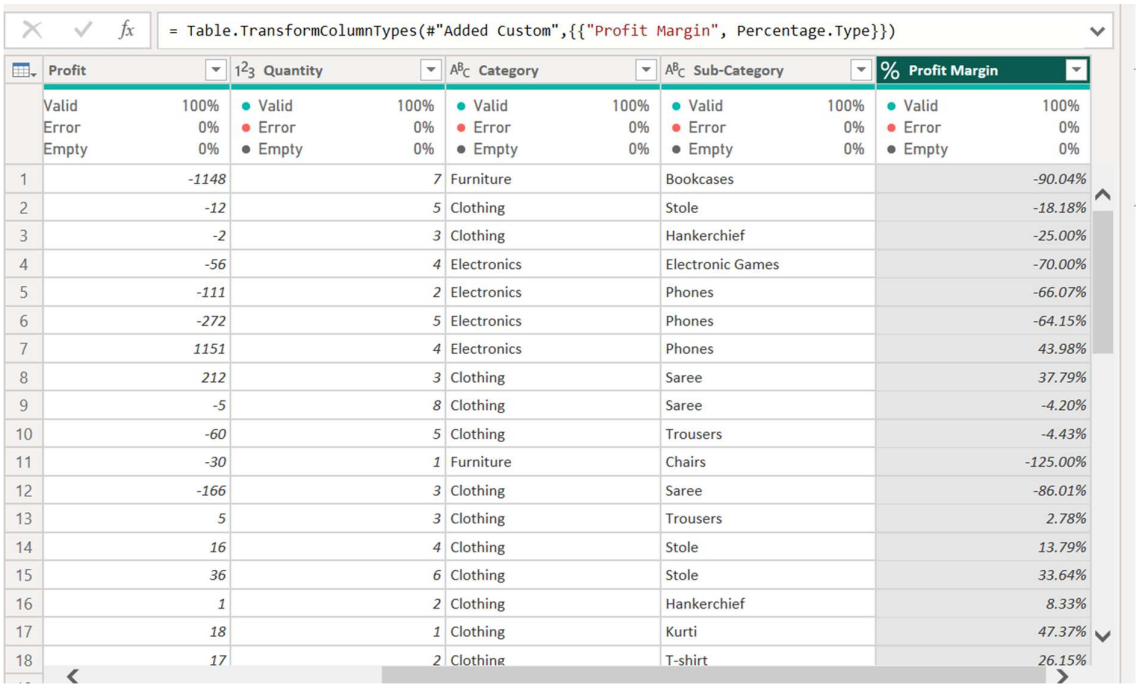
- Merge the City and State columns to create a new column named Location in the format: City, State



```
= Table.RenameColumns(#"Removed Columns",{{"Merged", "Location"}})
```

	Order Date	CustomerName	State	City	Location
	alid 100% rror 0% mpty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	01-04-2018	Bharat	Gujarat	Ahmedabad	Ahmedabad Gujarat
2	01-04-2018	Pearl	Maharashtra	Pune	Pune Maharashtra
3	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Bhopal Madhya Pradesh
4	03-04-2018	Divsha	Rajasthan	Jaipur	Jaipur Rajasthan
5	05-04-2018	Kasheen	West Bengal	Kolkata	Kolkata West Bengal
6	06-04-2018	Hazel	Karnataka	Bangalore	Bangalore Karnataka
7	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir	Kashmir Jammu and Kashmir
8	08-04-2018	Aarushi	Tamil Nadu	Chennai	Chennai Tamil Nadu
9	09-04-2018	Jitesh	Uttar Pradesh	Lucknow	Lucknow Uttar Pradesh
10	09-04-2018	Yogesh	Bihar	Patna	Patna Bihar
11	11-04-2018	Anita	Kerala	Thiruvananthapuram	Thiruvananthapuram Kerala
12	12-04-2018	Shrichand	Punjab	Chandigarh	Chandigarh Punjab
13	12-04-2018	Mukesh	Haryana	Chandigarh	Chandigarh Haryana
14	13-04-2018	Vandana	Himachal Pradesh	Simla	Simla Himachal Pradesh
15	15-04-2018	Bhavna	Sikkim	Gangtok	Gangtok Sikkim
16	15-04-2018	Kanak	Goa	Goa	Goa Goa
17	17-04-2018	Sagar	Nagaland	Kohima	Kohima Nagaland
18	18-04-2018	Maniu	Andhra Pradesh	Hyderabad	Hyderabad Andhra Pradesh

- Create a custom column named Profit Margin calculated as: Profit / Amount (expressed as a percentage)

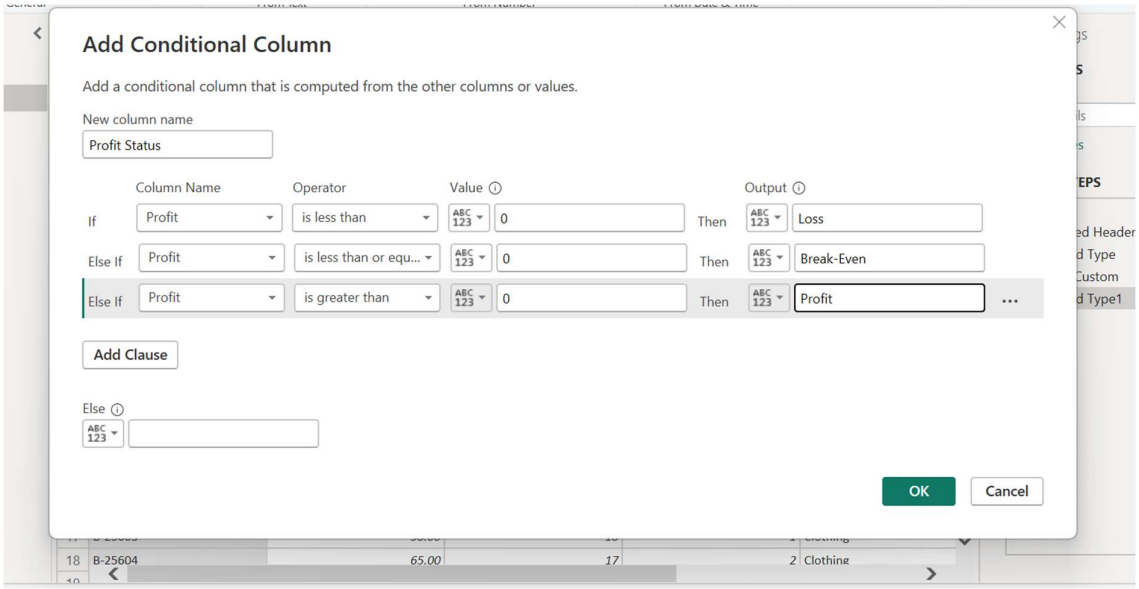


```
= Table.TransformColumnTypes(#"Added Custom",{{"Profit Margin", Percentage.Type}})
```

	Profit	Quantity	Category	Sub-Category	Profit Margin
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	-1148	7	Furniture	Bookcases	-90.04%
2	-12	5	Clothing	Stole	-18.18%
3	-2	3	Clothing	Hankerchief	-25.00%
4	-56	4	Electronics	Electronic Games	-70.00%
5	-111	2	Electronics	Phones	-66.07%
6	-272	5	Electronics	Phones	-64.15%
7	1151	4	Electronics	Phones	43.98%
8	212	3	Clothing	Saree	37.79%
9	-5	8	Clothing	Saree	-4.20%
10	-60	5	Clothing	Trousers	-4.43%
11	-30	1	Furniture	Chairs	-125.00%
12	-166	3	Clothing	Saree	-86.01%
13	5	3	Clothing	Trousers	2.78%
14	16	4	Clothing	Stole	13.79%
15	36	6	Clothing	Stole	33.64%
16	1	2	Clothing	Hankerchief	8.33%
17	18	1	Clothing	Kurti	47.37%
18	17	2	Clothing	T-shirt	26.15%

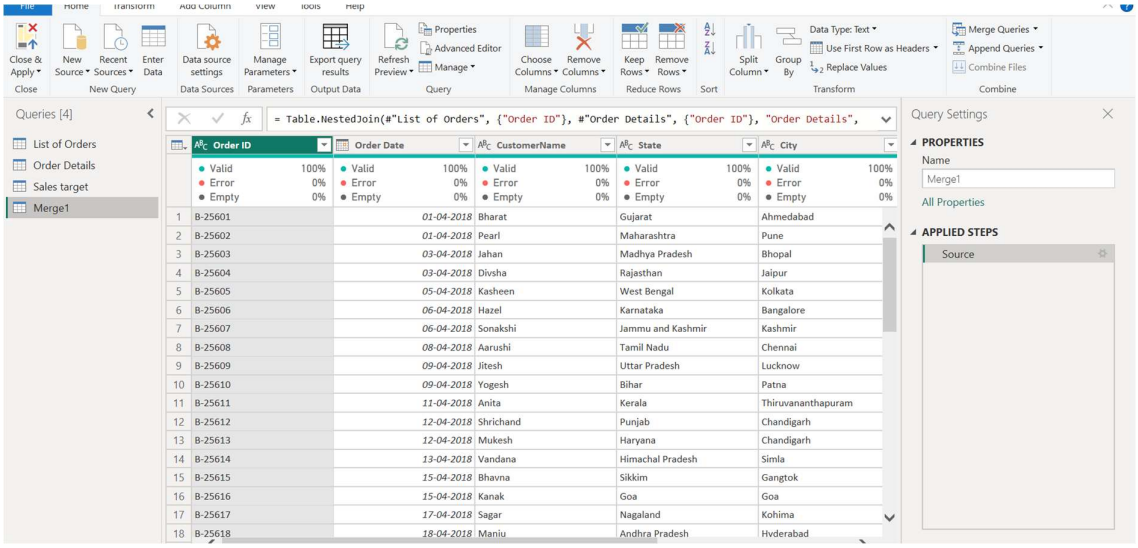
5.Conditional Column:

- Add a conditional column named Profit Status based on:
  - Profit < 0 → Loss
  - Profit = 0 → Break-Even
  - Profit > 0 → Profit



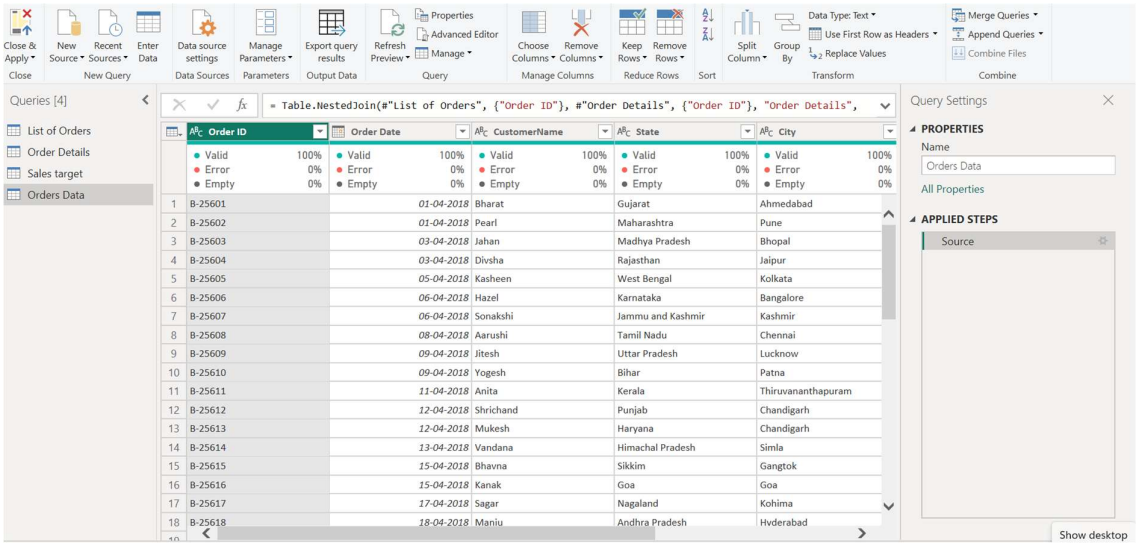
6.Merging Data:

- Merge the List of Orders and Order Details tables using Order ID.



	Order ID	Order Date	CustomerName	State	City
1	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
2	B-25602	01-04-2018	Pearl	Maharashtra	Pune
3	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
4	B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
5	B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
6	B-25606	06-04-2018	Hazel	Karnataka	Bangalore
7	B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
8	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
9	B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
10	B-25610	09-04-2018	Yogesh	Bihar	Patna
11	B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
12	B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
13	B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
14	B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
15	B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
16	B-25616	15-04-2018	Kanak	Goa	Goa
17	B-25617	17-04-2018	Sagar	Nagaland	Kohima
18	B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad

- Name the merged output table as Orders Data.

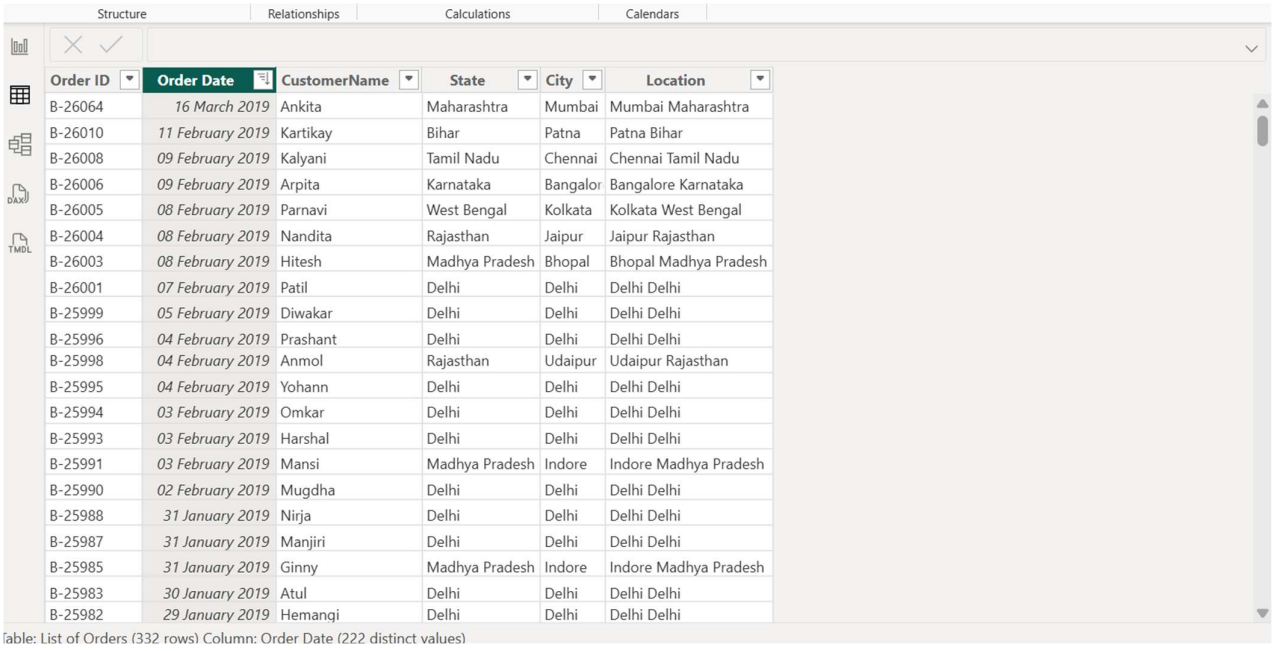


7.Handling Missing Data & Duplicate Data:

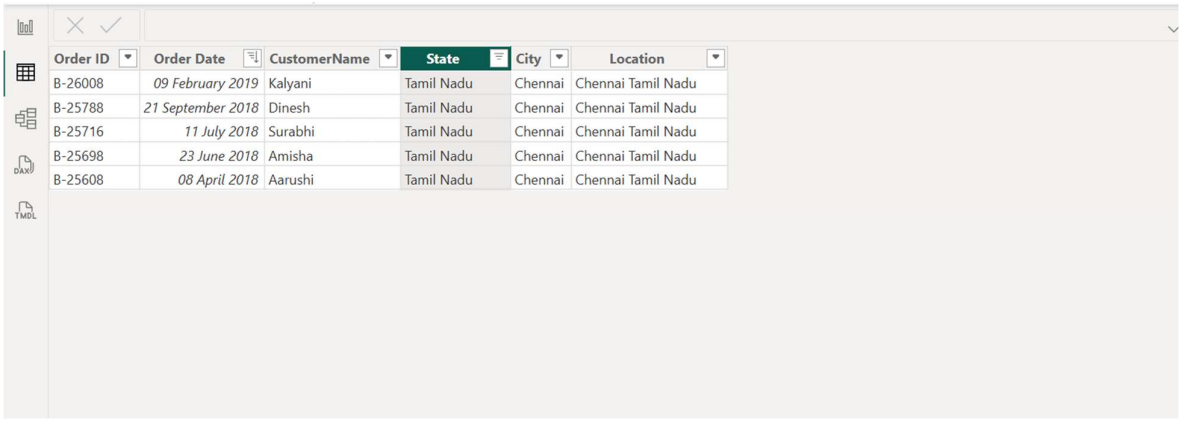
All the datas are properly stored.

8.Sorting and Filtering Data:

- Sort records in the Orders Data table by Order Date in descending order.



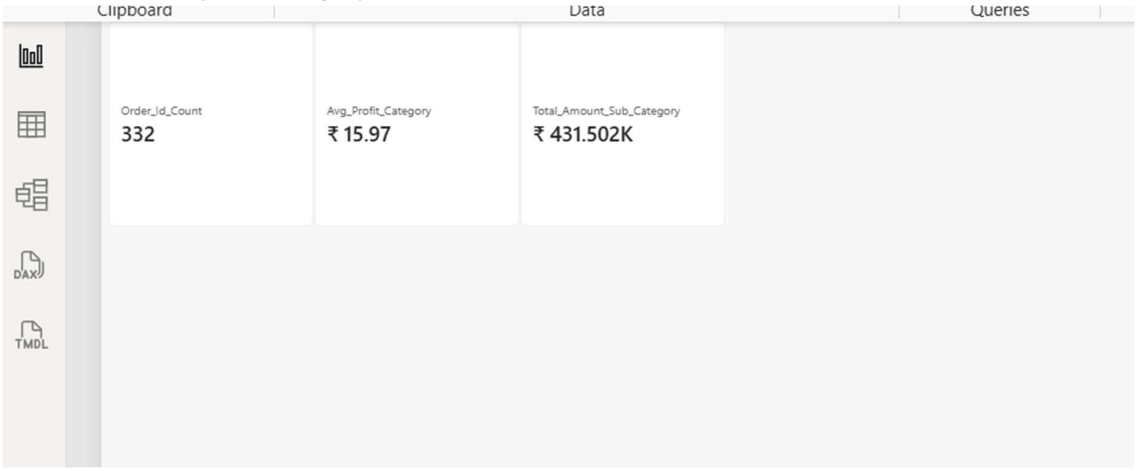
- Apply filters to analyze orders from a specific state (e.g., Tamil Nadu).



Order ID	Order Date	CustomerName	State	City	Location
B-26008	09 February 2019	Kalyani	Tamil Nadu	Chennai	Chennai Tamil Nadu
B-25788	21 September 2018	Dinesh	Tamil Nadu	Chennai	Chennai Tamil Nadu
B-25716	11 July 2018	Surabhi	Tamil Nadu	Chennai	Chennai Tamil Nadu
B-25698	23 June 2018	Amisha	Tamil Nadu	Chennai	Chennai Tamil Nadu
B-25608	08 April 2018	Aarushi	Tamil Nadu	Chennai	Chennai Tamil Nadu

9.Grouping and Aggregating Data:

- Duplicate the Order Details table and perform the following:
  - Count of Order ID -332
  - Average Profit by Category -15.97
  - Total Amount by Sub-Category-431.502K



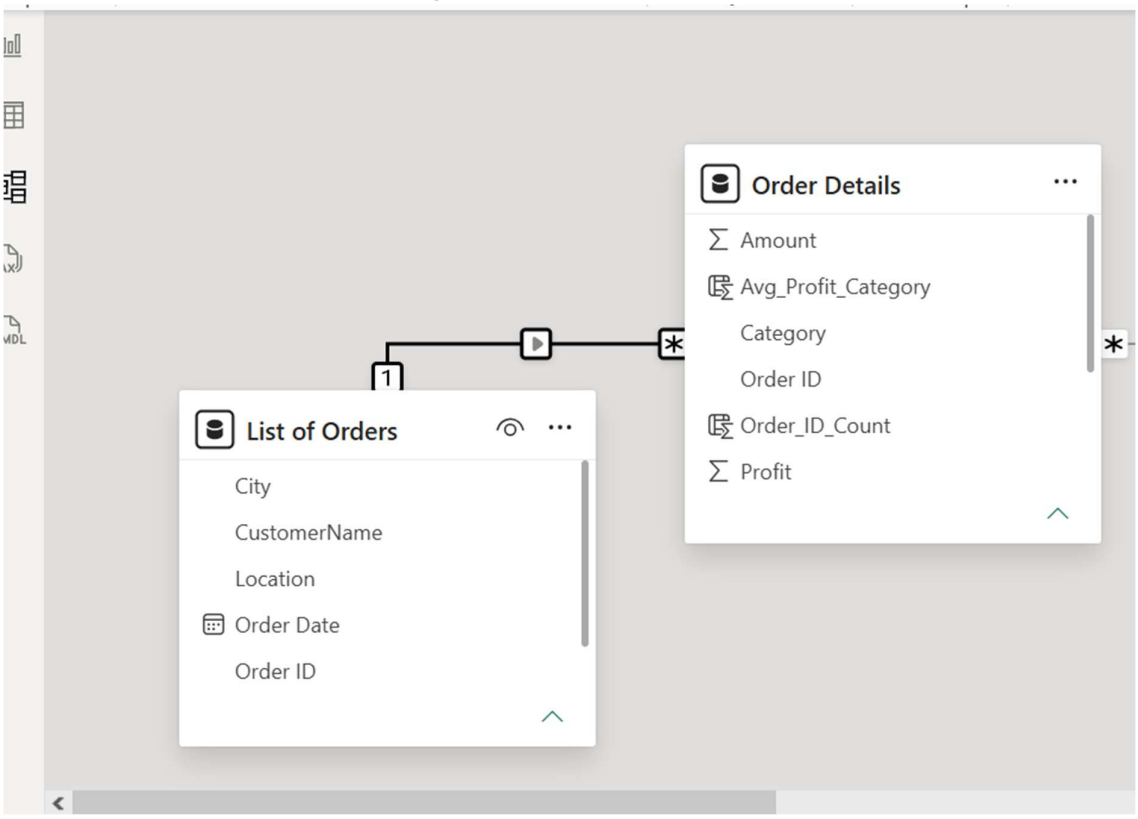
- Duplicate the Sales Target table and aggregate the total Target by Month of order date.



Monthly Target	Category
₹ 1,32,900	Furniture
₹ 1,74,000	Clothing
₹ 1,29,000	Electronics

10.Data Modelling:

- Establish a relationship between:
  ○ List of Orders and Order Details using Order ID



- Establish a relationship between:
  ○ Order Details and Sales Target using Category

